

AMS-02 Layer 0 Integration System

High-precision assembly tooling for silicon detector integration

The AMS-02 experiment operates aboard the International Space Station and uses a silicon tracker to measure charged cosmic rays. The Layer 0 upgrade introduces a new **2.6 m detector plane of 768 silicon sensors arranged into 72 detector ladders**, carried by lightweight carbon-composite QLCS quadrants — four per measurement plane.

2.6 m	768	72	200 μm	±100 μm
LAYER 0 DIAMETER	SILICON SENSORS	DETECTOR LADDERS	NOMINAL LADDER GAP	POSITIONING LIMIT

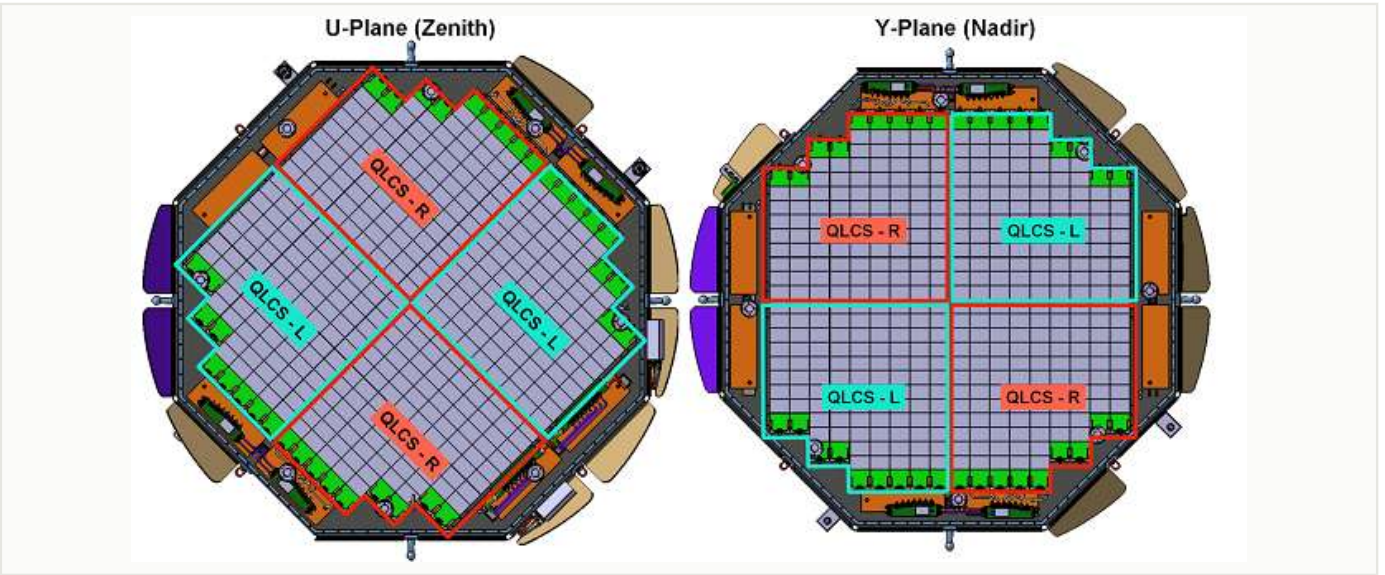


Fig. 1 – Layer 0 geometry. U-plane (zenith) and Y-plane (nadir), each divided into four QLCS quadrants carrying the silicon ladders.

WHAT HAS TO BE ASSEMBLED

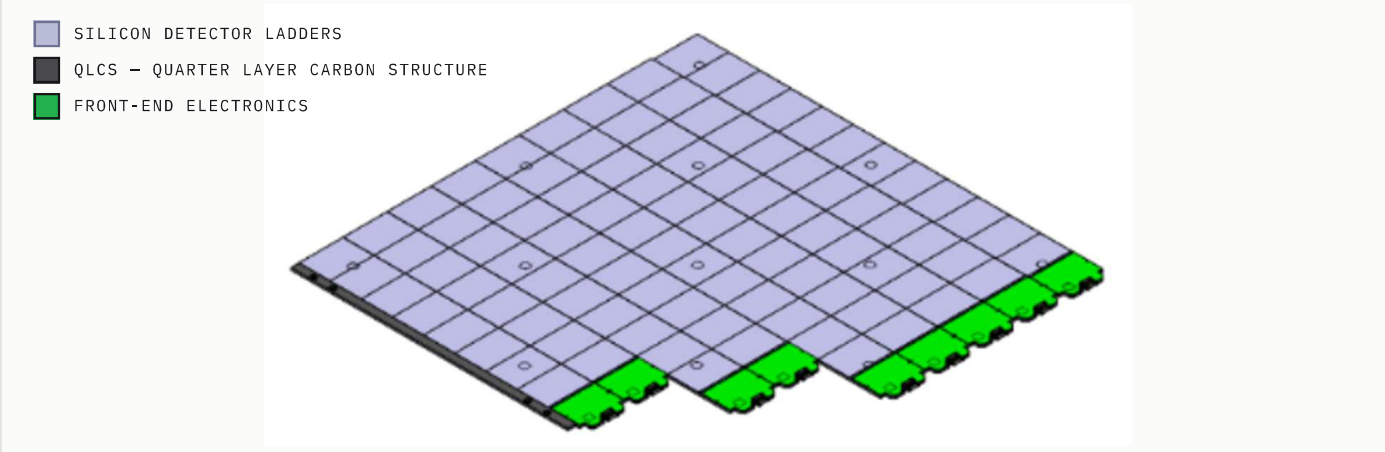


Fig. 2 – One QLCS (quarter ladder carbon structure) quadrant fully populated. Ladders are bonded side by side directly onto the carbon structure; the whole quadrant is a metre-scale, low-mass assembly.

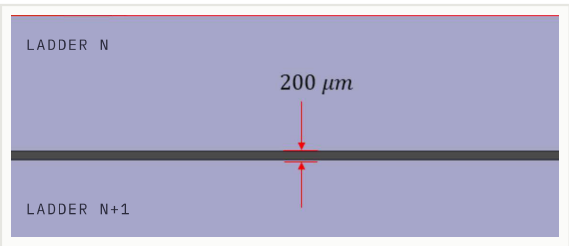


Fig. 3 – The interface between neighbouring ladders. 200 μm is the nominal physical gap, not the positioning tolerance.

ENGINEERING CHALLENGE

Fragile silicon detector ladders had to be positioned and adhesively bonded onto lightweight carbon-composite QLCS structures while maintaining a nominal **200 μm gap between adjacent ladders**. This required individual ladder positioning errors to remain within approximately **±100 μm**, while also controlling handling, structural deformation and the bonded geometry.

How the ladder meets the QLCS

The ladders are bonded directly onto the CFRP skin of the lightweight QLCS sandwich structure. The integration tooling therefore had to control both in-plane alignment and the vertical geometry of the bonded interface.

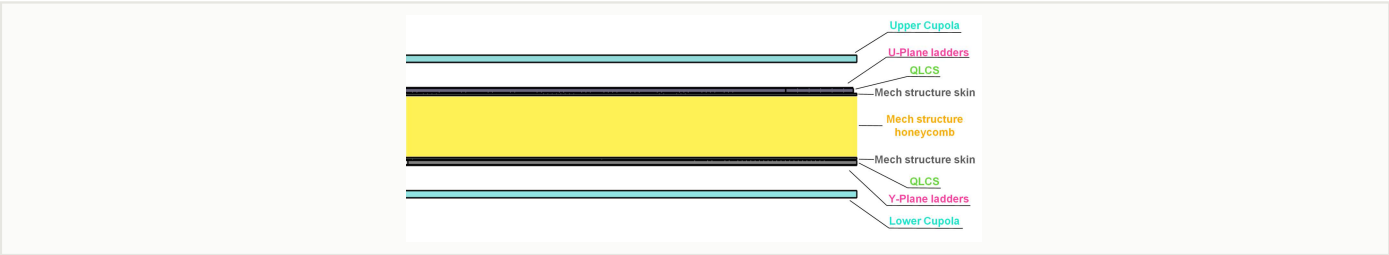


Fig. 4 – Section through Layer 0: both measurement planes sit on a common lightweight sandwich structure between the upper and lower cupolas.

LADDER-TO-QLCS BONDED STACK

Section through the bonded interface

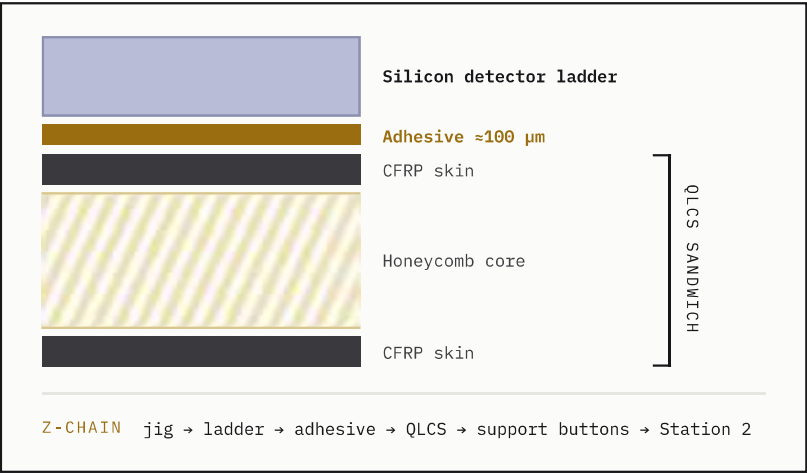


Fig. 5 – The bonded interface. Bond-line thickness is set by the relative heights of jig, ladder and QLCS, so Station 2 controls a vertical chain as well as an in-plane position.

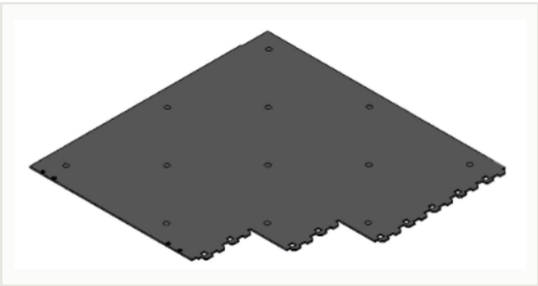
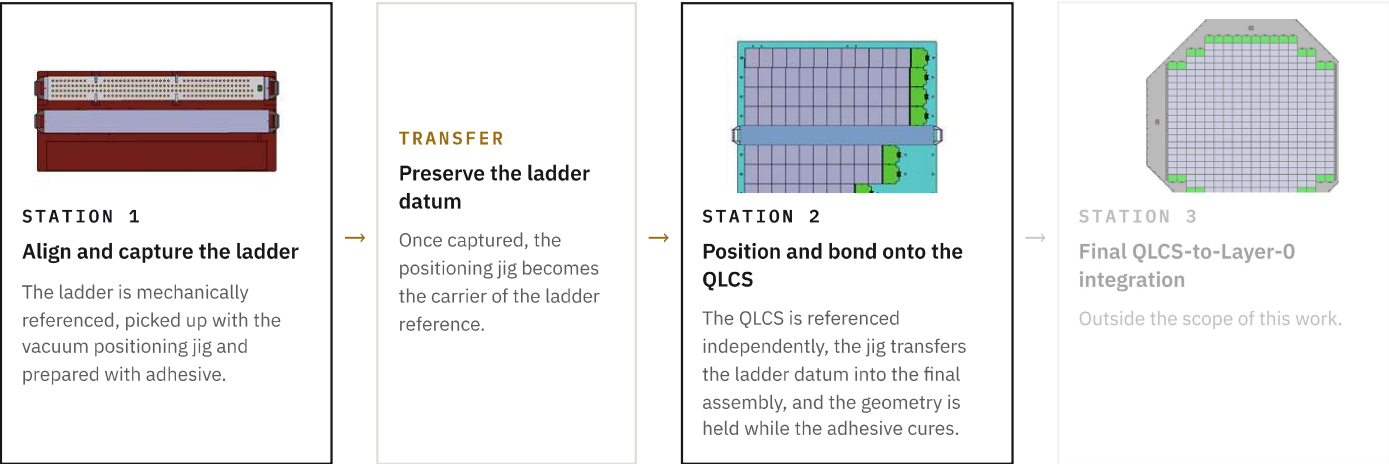


Fig. 6 – The bare QLCS: a CFRP-honeycomb sandwich panel, stiff for its mass and the substrate for every ladder in a quadrant.

Because the substrate is thin and compliant, both the adhesive thickness and the flatness of the panel during integration had to be held by the tooling rather than by the parts themselves.

INTEGRATION SEQUENCE



The positioning jig carries the ladder reference from Station 1 to Station 2 – the two stations are one continuous datum-transfer architecture, not two independent fixtures.

MY CONTRIBUTION

Mechanical design of Stations 1 and 2 of the Layer 0 integration system, covering ladder alignment, vacuum handling, adhesive preparation, QLCS positioning and precision ladder-to-QLCS integration.

Establish and preserve the ladder datum

Station 1 aligns the ladder against mechanical references, captures that alignment with a vacuum positioning jig, and prepares the bonding surface without the ladder ever losing its reference.

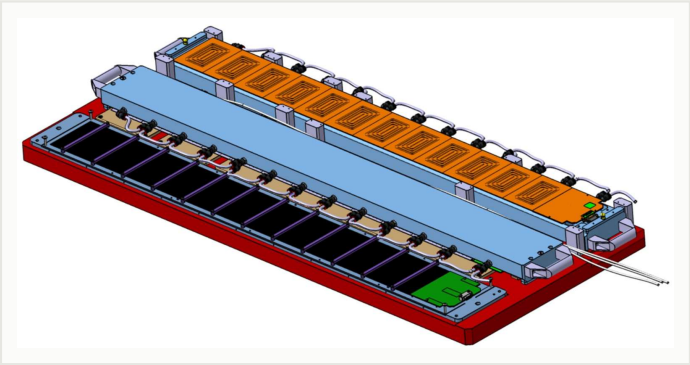


Fig. 7 – Station 1 carrying an 8-sensor ladder, with the positioning jig seated in its working configurations.

01

Align

The ladder leaves its transport container and is positioned against dedicated alignment features.

02

Capture

The jig is lowered onto it and retains it on distributed vacuum pads — no mechanical clamping onto the detector.

03

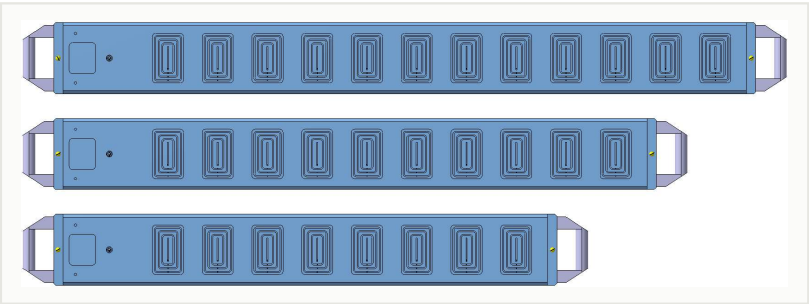
Preserve the datum

The jig becomes the carrier of the reference; spherical-ended pins hand it to the next station.

04

Apply adhesive

The jig is inverted to expose the bonding face and adhesive is laid through a bonding mask.



JIG FAMILY • L12 / L10 / L8

Different ladder lengths required dedicated jig geometries while preserving the same locating and vacuum-handling architecture.

Fig. 8 – The three positioning jigs shown at the same scale.

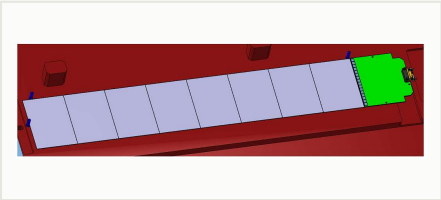


Fig. 9 – Locating features in the alignment configuration.

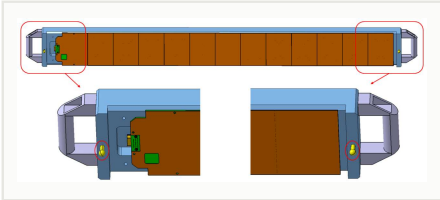


Fig. 10 – Jig underside: distributed vacuum pads.

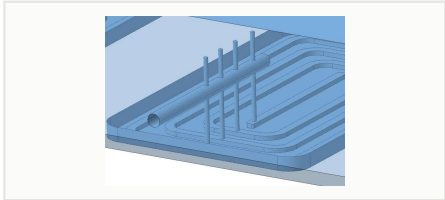


Fig. 11 – Internal vacuum channels machined into the jig body.

STRUCTURAL SIMULATION – JIG 12

Static structural • total deformation

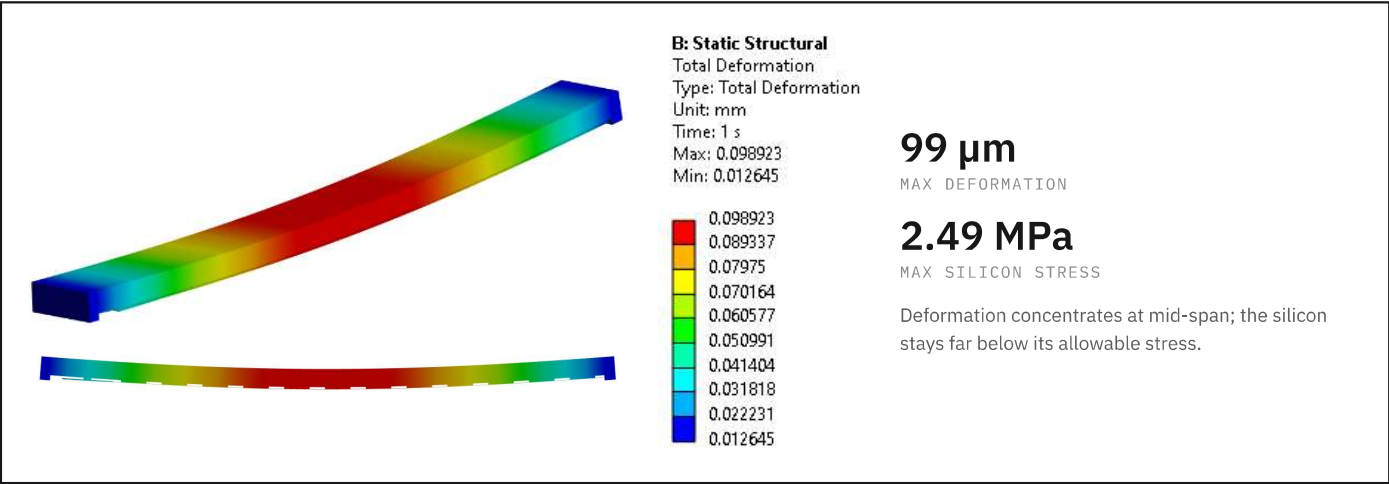


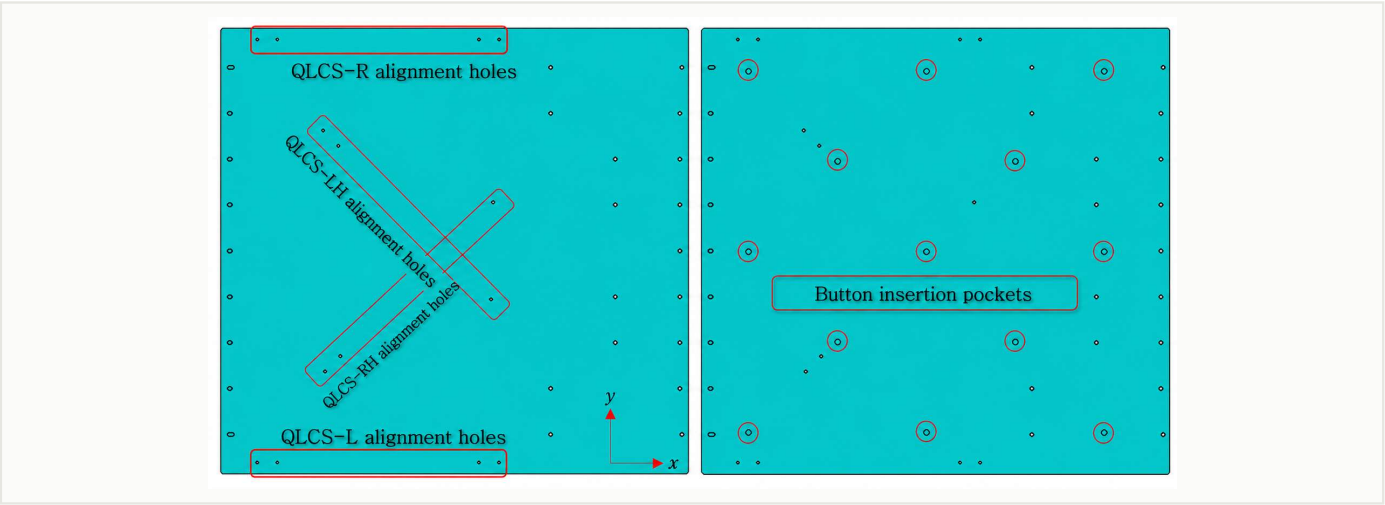
Fig. 12 – FEM of the jig configured for 12 sensors, loaded by the ladder under vacuum.

OUTPUT OF STATION 1

Ladder aligned, captured by the vacuum jig and prepared with adhesive while preserving its positional reference.

Transfer the datum and control the bonded geometry

Station 2 is a 1100 × 1200 mm bonding fixture on a 20 mm plate, sized as a compromise between stiffness and handling mass. It controls four things at once: where the QLCS sits, where the ladder sits, how thick the adhesive layer is, and how the assembly is held while it cures.



X-Y REFERENCE Alignment holes for each of the four QLCS. **Z REFERENCE** Pockets receiving the support buttons.

Fig. 13 – The Station 2 plate carries both QLCS reference systems: alignment holes fix the quadrant in the plane, button pockets fix it in height.

QLCS DATUM

- X-Y** Precision pins, holes and slots establish the in-plane position without unnecessarily over-constraining the structure.
- Z** Support buttons establish the vertical datum, distribute the QLCS self-weight and limit out-of-plane deformation during ladder integration.

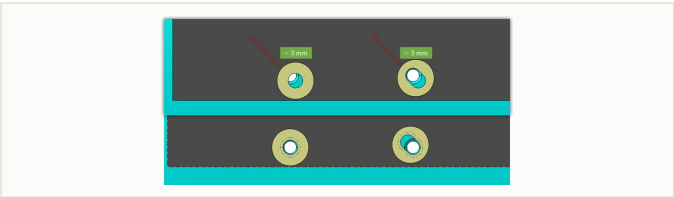


Fig. 14 – Hole and slot: the pin is fully constrained at one end and free along the slot at the other.

CONTROLLED BONDED GEOMETRY

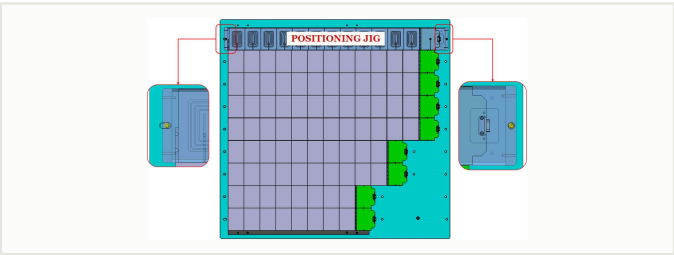


Fig. 15 – The jig seated on Station 2, holding the ladder over the QLCS in its final position.

LADDER DATUM

- X-Y** The ladder arrives from the station 1 held under vacuum by positioning jig; locating the jig onto Station 2 transfers that reference into the final ladder-to-QLCS position.
- Z** Relative heights of jig, ladder and QLCS set the bonded geometry – a ~100 μm adhesive layer after integration.
- JIG** The jig locating elements repeat the Station 1 architecture, so the same interface receives the jig at both stations.

Jig → ladder → adhesive → QLCS → buttons → station

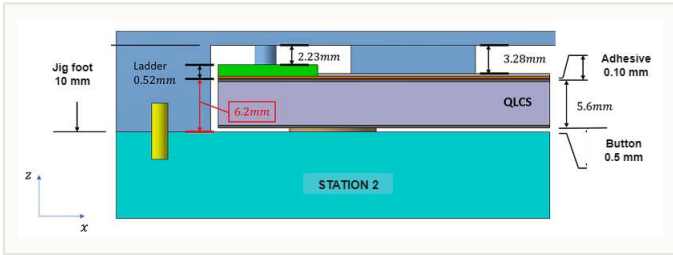


Fig. 16 – xz section of the final configuration: everything between the jig foot and the station plate is a controlled dimension, and the adhesive layer is the smallest of them.

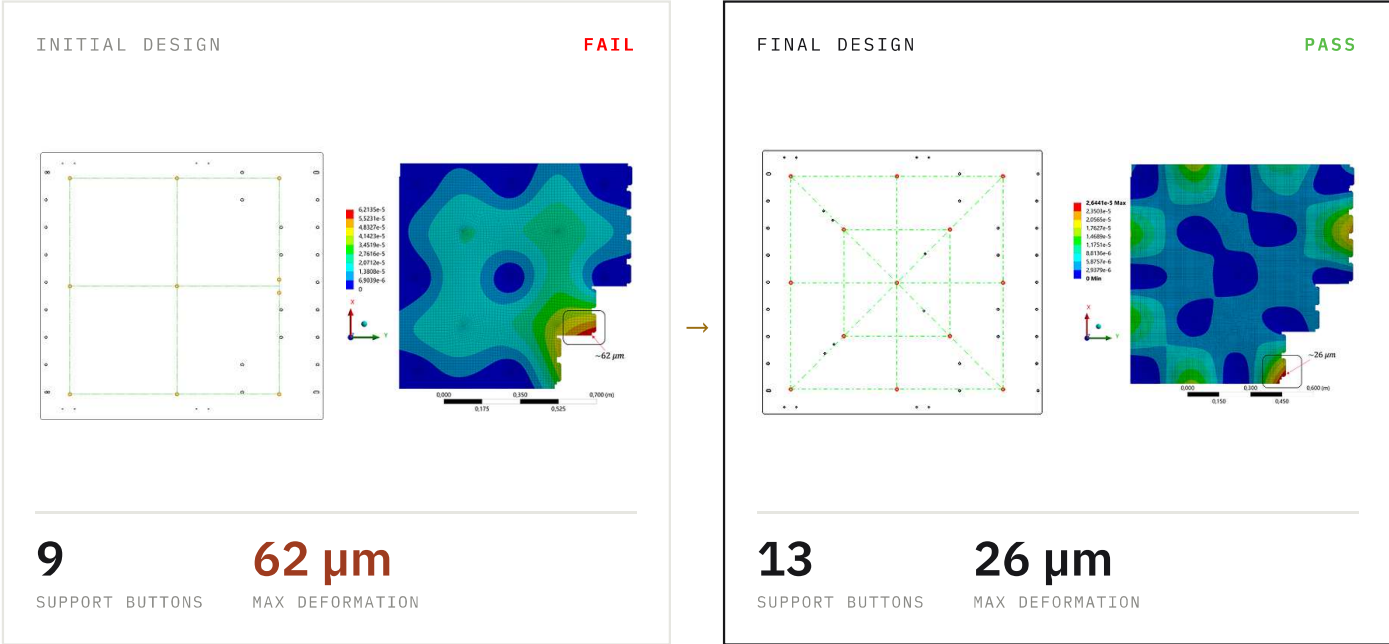
DESIGN DETAIL – SPHERICAL-ENDED LOCATING PIN

A metre-scale jig cannot be lowered perfectly vertically by hand. The spherical-ended lead-in reduces sensitivity to angular misalignment during initial engagement; once fully seated, the locating interface establishes the precision datum.

The architecture was analysed, not assumed

Two questions decided part of the design: whether the buttons support layout kept the QLCS flat enough during integration, and whether the accumulated fits and machining tolerances could still land a ladder inside its allowance.

QLCS SUPPORT OPTIMISATION Requirement < 50 µm out-of-plane



POSITIONING TOLERANCE

Nominal ladder gap	200 µm
Allowable deviation	100 µm
Calculated worst case	93 µm
Status	PASS

Shaft-hole and shaft-slot fits, machining and hole positional tolerances and QLCS rotation accumulated into a worst-case ladder displacement.

THE FIT

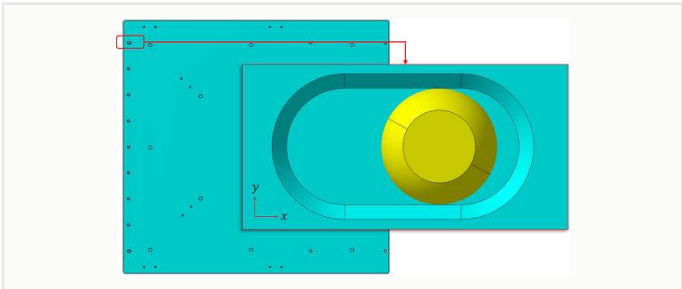


Fig. 17 – Shaft-slot coupling between pin and station: one of the fits carried into the stack-up.

RESULT

In the analysed design conditions, the integration architecture satisfies both the structural and the positioning requirements that govern Layer 0 assembly.

ENGINEERING INSIGHT

Stations 1 and 2 form a continuous datum-transfer architecture. Station 1 establishes and captures the ladder reference; Station 2 references the QLCS and transfers that datum into the final bonded assembly. Vacuum handling, hole-slot constraints, spherical locating features and the optimised support layout all serve the same objective: preserving the required alignment through handling, transfer and adhesive curing.

From CAD to hardware

The designed tooling was manufactured and assembled, and Layer 0 integration was carried out with it in the clean room.

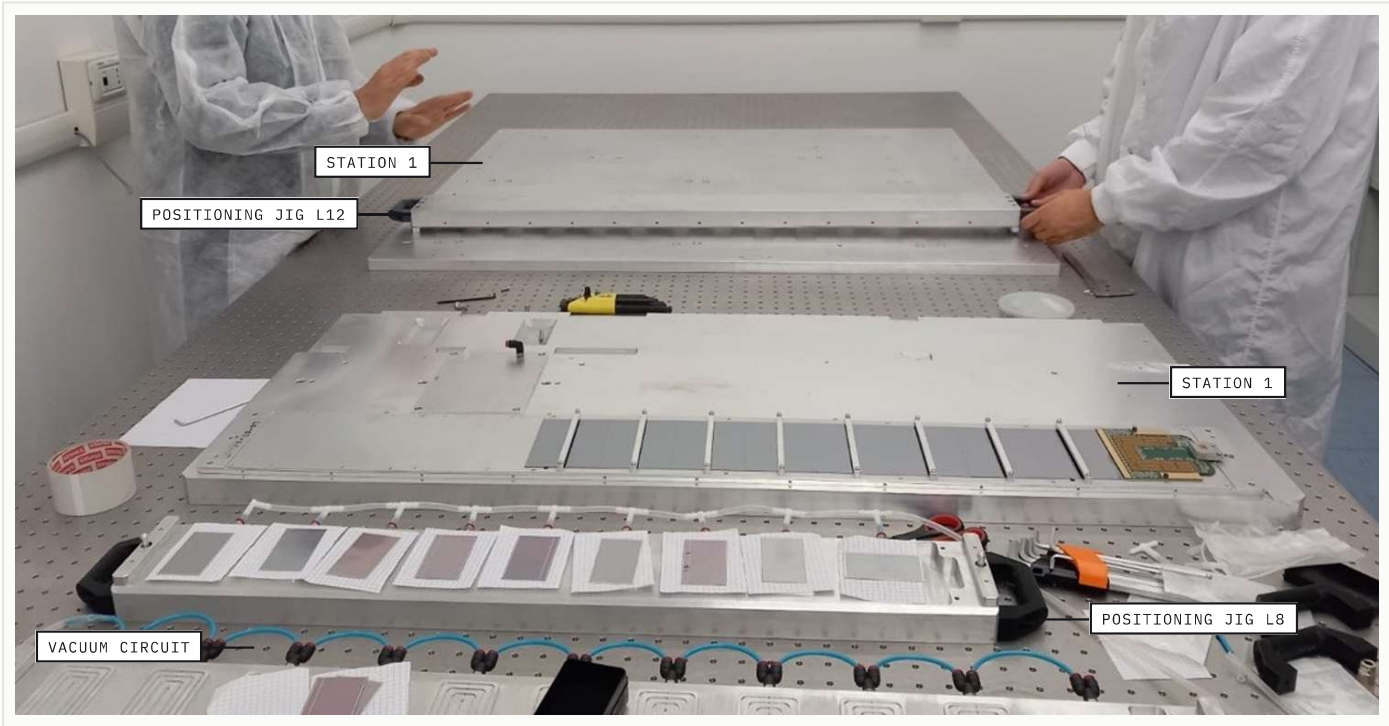


Fig. 18 – The integration tooling in use: machined station plates, ladders held on the positioning tooling and the distributed vacuum circuit connected.

DESIGN → MANUFACTURED HARDWARE

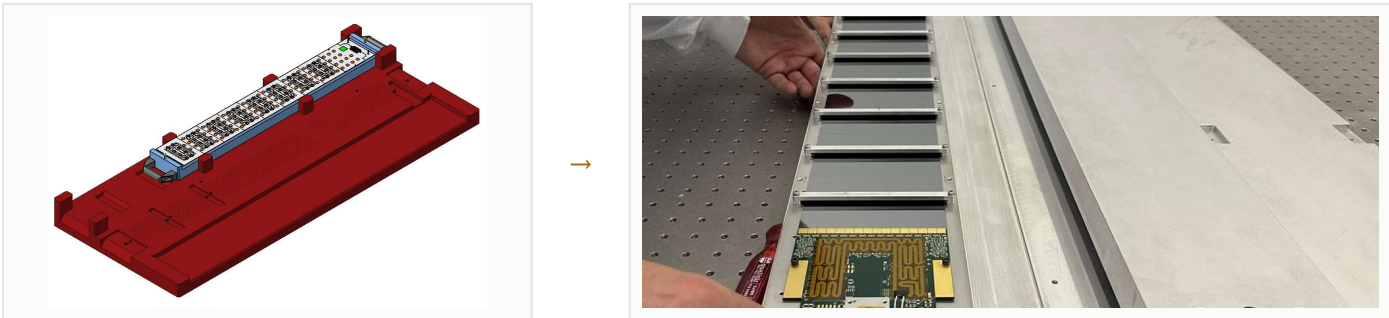


Fig. 19 – Station 1: the CAD assembly and the manufactured station in use.

STATION 1

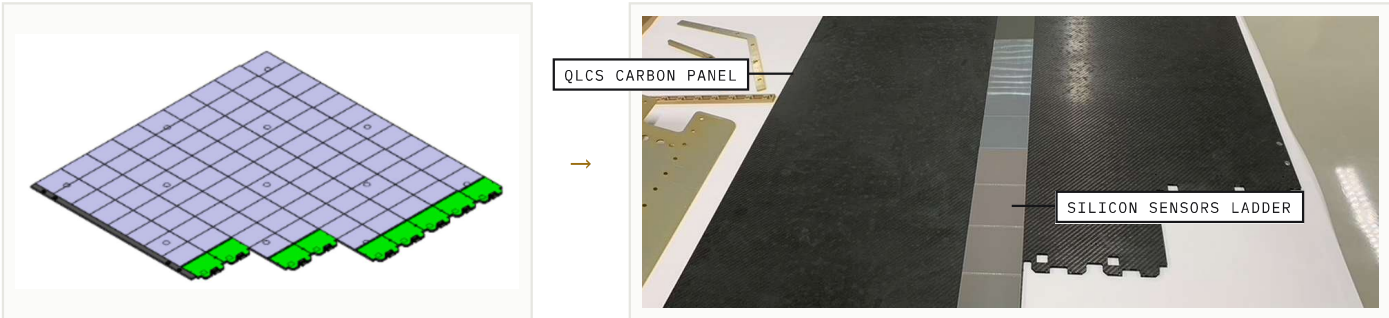


Fig. 20 – The QLCS: the designed quadrant and the manufactured carbon panel with a ladder on it.

QLCS AND LADDER

The final tooling translated the analysed datum architecture into manufacturable precision hardware for Layer 0 detector integration.